What is a Tuple in Python?

A **tuple** is an **ordered**, **immutable** (unchangeable) **collection** of items in Python.
It can store **multiple items** of **different data types** (like strings, numbers, lists, etc.).

🧠 **Syntax**

tuple_name = (item1, item2, item3)

Example:

my_tuple = (10, 20, 30, 40)

print(my_tuple)

Output:

(10, 20, 30, 40)

🔹 **Characteristics of Tuples**

Property	Description
Ordered	Items have a fixed order (index-based).
Immutable	Cannot change items after creation.
Allows duplicates	Yes, repeated values are allowed.
Can contain multiple types	Example: (1, "hello", 3.5)

🔹 **Tuple Example with Mixed Data Types**

student = ("John", 21, "Computer Science", 9.2)

print(student)

Output:

('John', 21, 'Computer Science', 9.2)

🔹 **Accessing Tuple Elements (Indexing)**

```
fruits = ("apple", "banana", "cherry")


print(fruits[0])   # apple
print(fruits[2])   # cherry
```

🔹 **Negative Indexing**

```
fruits = ("apple", "banana", "cherry")
print(fruits[-1])  # cherry (last element)
```

🔹 **Slicing Tuples**

```
numbers = (10, 20, 30, 40, 50)
print(numbers[1:4])   # (20, 30, 40)
print(numbers[:3])    # (10, 20, 30)
print(numbers[::2])   # (10, 30, 50)
```

🔹 **Tuple Immutability (Can’t Change Items)**

```
colors = ("red", "green", "blue")
colors[0] = "yellow"   # ❌ This will cause an error
```

Output:

```
TypeError: 'tuple' object does not support item assignment
```

🔹 **Looping Through a Tuple**

```
numbers = (10, 20, 30)
for num in numbers:
```

```
print(num)
```

🔹 Tuple Length

```
colors = ("red", "green", "blue")
print(len(colors))   # 3
```

🔹 Tuple with One Item

Be careful — a single item **must have a comma** after it!

```
single_tuple = ("apple",)
print(type(single_tuple))  # <class 'tuple'>

not_tuple = ("apple")
print(type(not_tuple))     # <class 'str'>
```

🔹 Tuple Operations

Operation	Example	Result
Concatenation	(1, 2) + (3, 4)	(1, 2, 3, 4)
Repetition	(5, 6) * 2	(5, 6, 5, 6)
Membership	3 in (1, 2, 3)	True

🔹 Nested Tuple

```
nested = (1, 2, (3, 4, 5))
print(nested[2])     # (3, 4, 5)
print(nested[2][1])  # 4
```

🔹 Tuple Methods

Tuples have only **two** built-in methods:

Method	Description	Example
.count(x)	Returns how many times x appears	(1, 2, 2, 3).count(2) → 2
.index(x)	Returns index of first occurrence	(1, 2, 3).index(3) → 2

🔹 Tuple Packing and Unpacking

```
# Packing
student = ("John", 21, "CS")

# Unpacking
name, age, course = student

print(name)   # John
print(age)    # 21
print(course) # CS
```

🧠 Summary

Concept	Example
Create	t = (1, 2, 3)
Access	t[0]
Immutable	❌ t[0] = 5
Nested	(1, (2, 3))
Methods	count(), index()

Concept	Example
Unpack	a, b = (10, 20)